

PROJECT SOL-SENTINEL

System Design Equations

Member C & D | Power, Thermal & Communications

KEY ASSUMPTION — Ion thruster (BIT-3) and Iris transmitter **NEVER** operate simultaneously. Solar array is sized to the worst-case single operational mode: Mode 3 Station-keeping (thruster ON, TX OFF, P = 71.9 W).

1. SOLAR ARRAY EFFICIENCY CHAIN (η_{total})

Every watt of sunlight hitting the panels loses energy at five sequential stages before becoming usable bus power. Each factor multiplies together:

Factor	Equation / Value	Result	Confidence	Source
η_{cell} GaAs BOL efficiency	$\eta_{\text{cell}} = 0.300$	0.300	High ☒	BCT Digital Catalog p.46: '30 percent efficient cells'
η_{temp} Temperature derating	$\eta_{\text{temp}} = 1 - (T_{\text{op}} - T_{\text{std}}) \times \gamma = 1 - (70 - 28) \times 0.002 = 1 - 0.084$	0.916	Medium	GaAs $\gamma = 0.002/^{\circ}\text{C}$ standard. $T_{\text{op}} = 70^{\circ}\text{C}$ (no eclipse at L4). EnduroSat 6U datasheet; Spectrolab.
η_{rad} Radiation degradation	$\eta_{\text{rad}} = (1 - r)^n = (1 - 0.03)^2 = (0.97)^2$	0.9409	Medium	$r = 3\%/yr$ for GaAs deep space. JPL MarCO heritage (Levine et al.); NASA SMAD.
η_{MPPT} MPPT efficiency	$\eta_{\text{MPPT}} = 1.000$ (assumed ideal)	1.000	Assumed ☐	No numeric figure in STARBUCK-NANO datasheet. Real ~93–97%. +7% power contingency budgeted.
η_{wiring} Wiring & mismatch	$\eta_{\text{wiring}} = 1.000$ (assumed ideal)	1.000	Assumed ☐	No datasheet source. Typical 2–5% loss. +5% power margin recommended.
η_{TOTAL} Combined	$\eta_{\text{total}} = \eta_{\text{cell}} \times \eta_{\text{temp}} \times \eta_{\text{rad}} \times \eta_{\text{MPPT}} \times \eta_{\text{wiring}} = 0.300 \times 0.916 \times 0.9409 \times 1.0 \times 1.0$	0.2586	Medium	Product of all five factors above.

2. POWER BUDGET — P_{consumed} per Operational Mode

Total bus power consumed is the sum of all active subsystems. The design driver is the mode with highest demand.

2.1 General Equation

$$P_{\text{consumed}} = P_{\text{payload}} + P_{\text{comms}} + P_{\text{ADCS}} + P_{\text{CDH}} + P_{\text{EPS}} + P_{\text{thermal}} + P_{\text{propulsion}}$$

2.2 Mode Totals

Subsystem	Mode 1: Science (W)	Mode 2: Downlink (W)	Mode 3: Station-keeping (W)
SIS + VHM + MERiT + DHU (Payload)	2.2	2.2	2.2
Iris Transponder (Comms)	6.4 (RX only)	12.75 (full TX)	6.4 (RX only)
XACT-100 (ADCS)	4.0	4.0	4.0
OBC + Flash (CDH)	0.7	0.7	0.7
STARBUCK + OPTIMUS (EPS)	2.1	2.1	2.1
Kapton heaters (Thermal)	0.5	0.5	0.5
BIT-3 Ion Thruster (Propulsion)	0.0 (OFF)	0.0 (OFF)	56.0 (firing)
TOTAL P_consumed	15.9 W	22.25 W	71.9 W ← DESIGN DRIVER

2.3 Simultaneous Operation Constraint

$$P_{\text{design}} = \max(P_{\text{downlink}}, P_{\text{stationkeep}}) \quad \text{NOT} \quad P_{\text{TX}} + P_{\text{thruster}} \quad P_{\text{design}} = \max(22.25, 71.9) = 71.9 \text{ W}$$

3. SOLAR ARRAY AREA (A_sa)

3.1 Core Formula

$$A_{\text{sa}} = P_{\text{required}} / (S_0 \times \eta_{\text{total}})$$

3.2 Substituting Values

Parameter	Value	Unit	Source
S ₀ Solar flux at L4	1361	W/m ²	Standard value at 1 AU — NASA/NOAA solar irradiance tables
P _{required} (design driver)	71.9	W	Mode 3 Station-keeping total — design driver
η _{total} Combined efficiency	0.2586	—	From Section 1: 0.300 × 0.916 × 0.9409 × 1.0 × 1.0
A _{sa} = P / (S ₀ ×	= 71.9 / (1361 ×	m ²	Core formula applied directly

η_{total})	0.2586)		
A_sa (no margin)	0.204	m ²	= 2,040 cm ²
System margin (×1.20)	× 1.20	—	Standard 20% engineering margin
A_sa_FINAL (with margin)	0.245 m²	m ²	= 2,450 cm ² ≈ two EnduroSat 6U deployable panels

4. INTERNAL HEAT GENERATION (P_dissipated)

4.1 Core Formula

$P_{\text{dissipated}} = P_{\text{consumed}} - P_{\text{useful_out}}$ where $P_{\text{useful_out}}$ = RF power radiated as signal (1 W from Iris during TX only) All other consumed power becomes heat inside the spacecraft.

4.2 Results by Mode

Mode	P_consumed (W)	P_useful_out (W)	P_dissipated (W)	Physical Meaning
Mode 1 — Science (RX, no thrust)	15.9	0.0	15.9 W	All consumed power becomes internal heat.
Mode 2 — Downlink (full TX, no thrust)	22.25	1.0 (Iris RF)	21.25 W	1 W exits as radio signal. Rest is heat.
Mode 3 — Station-keeping (thrust, RX)	71.9	0.0	71.9 W ← DRIVER	Thermal radiator must reject 71.9 W.

5. THERMAL RADIATOR SIZING (A_rad → M_thm, V_thm)

5.1 Governing Equation (Stefan-Boltzmann from Fan Shi lecture)

At thermal equilibrium, heat rejected by the radiator must equal internal heat generated:

$Q_{\text{reject}} = P_{\text{dissipated}} = \epsilon \times \sigma \times A_{\text{rad}} \times T_{\text{max}}^4 - \alpha \times S_0 \times A_{\text{rad}}$
Rearranging to solve for A_rad: $A_{\text{rad}} = P_{\text{dissipated}} / (\epsilon \times \sigma \times T_{\text{max}}^4 - \alpha \times S_0)$

5.2 Input Values

Parameter	Value	Unit	Source / Justification
ϵ (OSR emissivity)	0.80	—	Fan Shi lecture table (slide 10): Silver-coated Teflon / OSR. Confirmed radiator selection.
α (OSR absorptivity)	0.08	—	Fan Shi lecture table (slide 10): Silver-coated Teflon $\alpha = 0.08$.

σ Stefan-Boltzmann constant	5.6696×10^{-8}	W/m ² /K ⁴	Fan Shi lecture (slide 14): $\sigma = 5.6696 \times 10^{-8}$ W/m ² /K
T_max (most restrictive component)	323 K (+50°C)	K	OPTIMUS-30 datasheet: operating temp limit +50°C — governs thermal design.
S ₀ Solar flux at L4	1361	W/m ²	Standard 1 AU irradiance.
P_dissipated (design driver)	71.9	W	Mode 3 Station-keeping — from Section 4.

5.3 Calculation

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Denominator = ( $\epsilon \times \sigma \times T_{\text{max}}^4$ ) - ( $\alpha \times S_0$ ) = ( $0.80 \times 5.6696 \times 10^{-8} \times 323^4$ )
- ( $0.08 \times 1361$ ) = ( $0.80 \times 5.6696 \times 10^{-8} \times 1.087 \times 10^{10}$ ) - 108.9
= 492.8 - 108.9 = 383.9 W/m2  A_rad = 71.9 / 383.9 = 0.187 m2
A_rad_FINAL (with 20% margin) = 0.187 × 1.20 = 0.225 m2

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6. RADIATION PRESSURE COEFFICIENT ($\rho_{\text{eff}} \rightarrow$ Orbit block)

ρ_{eff} (also written C_r) is the effective radiation pressure coefficient. It feeds from Thermal into the Orbit block so Member A can compute the SRP perturbation force on Sol-Sentinel at L4.

6.1 Coefficient Definition

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C_r = 1 +  $\rho_{\text{surface weighted}}$   Black body (fully absorbing): C_r = 1.0  White body
(fully reflecting): C_r = 2.0  Sol-Sentinel mixed coating: C_r = 1.820
(area-weighted - calculated below)  Face-by-face area-weighted calculation:
 $\rho_{\text{weighted}} = \Sigma(\rho_i \times A_i) / A_{\text{total}}$   Sun face (GaAs,  $\alpha=0.75$ ):  $\rho=0.250$ ,
A=0.02263 m2 →  $\rho \times A = 0.005658$   Back (OSR,  $\alpha=0.08$ ):  $\rho=0.920$ , A=0.02263
m2 →  $\rho \times A = 0.020820$   Top (OSR,  $\alpha=0.08$ ):  $\rho=0.920$ , A=0.07706 m2 →
 $\rho \times A = 0.070895$   Bottom (OSR,  $\alpha=0.08$ ):  $\rho=0.920$ , A=0.07706 m2 →  $\rho \times A =$ 
0.070895  Left side (white,  $\alpha=0.248$ ):  $\rho=0.752$ , A=0.03405 m2 →  $\rho \times A = 0.025606$ 
Right side (white,  $\alpha=0.248$ ):  $\rho=0.752$ , A=0.03405 m2 →  $\rho \times A = 0.025606$    $\Sigma(\rho \times A) =$ 
0.219480 | A_total = 0.26748 m2   $\rho_{\text{weighted}} = 0.219480 / 0.26748 = 0.820$ 
C_r = 1 + 0.820 = 1.820   $\alpha$  values from Fan Shi lecture slide 10 materials table.
Note: Previous estimate of C_r ≈ 1.5 was a rough midpoint - corrected to 1.820.

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6.2 SRP Force on Spacecraft (for Member A's CR3BP model)

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F_SRP = (S_0 / c) × A_projected × C_r  where:  S_0 = 1361 W/m2  c
= 3 × 108 m/s (speed of light)  A_projected = A_sa_FINAL = 0.245 m2 (solar
array dominates projected area)  C_r = 1.820 (area-weighted - corrected
from earlier estimate of 1.5)  F_SRP = (1361 / 3×108) × 0.245 × 1.820  F_SRP ≈ 2.02
× 10-6 N  Note: Previous estimate using C_r = 1.5 gave F_SRP = 1.67 × 10-6 N.
Corrected C_r = 1.820 increases SRP force by 21% - Member A must use the
corrected value in the CR3BP simulation to avoid underestimating drift.

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7. PROPULSION ($\Delta V \rightarrow M_{\text{prop}}, V_{\text{prop}}$) — Awaiting Member A

These equations are Member B's deliverable. They depend on ΔV from Member A's CR3BP orbital simulation. The equations are provided here for completeness and to show the coupling to your A_sa output.

7.1 Tsiolkovsky Rocket Equation (propellant mass)

$$m_{prop} = m_{spacecraft} \times (1 - e^{(-\Delta V / (Isp \times g_0))})$$

where: $m_{spacecraft}$ = total wet mass (from MEL) [kg] ΔV = station-keeping delta-V from Member A orbit simulation [m/s] Isp = 2150 s (BIT-3 specific impulse – Busek datasheet) g_0 = 9.807 m/s²

7.2 Propellant Tank Volume

$$V_{prop} = m_{prop} / \rho_{iodine}$$

where: $\rho_{iodine} \approx 4940 \text{ kg/m}^3$ (solid iodine – much denser than xenon, which is why BIT-3 uses it to minimise tank volume) Δ TBD – awaiting ΔV from Member A

8. SUMMARY — All Equations at a Glance

Diagram Relation	Equation	Result	Status
Comms → P_tx	Link budget → Iris datasheet confirmed	12.75 W / 6.4 W	✓ Done
Power → η_total	$\eta_{cell} \times \eta_{temp} \times \eta_{rad} \times \eta_{MPPT} \times \eta_{wiring}$	0.2586 (25.86%)	✓ Done
η_temp	$1 - (T_{op} - T_{std}) \times \gamma = 1 - (70-28) \times 0.002$	0.916	✓ Done
η_rad	$(1 - 0.03)^2$	0.9409	✓ Done
Power → P_consumed	ΣP_subsystems per mode (thruster & TX never simultaneous)	71.9 W (Mode 3)	✓ Done
Power → A_sa	$P_{required} / (S_0 \times \eta_{total}) = 71.9 / (1361 \times 0.2586)$	0.245 m² (w/ margin)	✓ Done
Power → P_dissipated	$P_{consumed} - P_{useful_out} = 71.9 - 0$	71.9 W (Mode 3)	✓ Done
Thermal → A_rad (M_thm)	$P_{diss} / (\epsilon \times \sigma \times T^4 - \alpha \times S_0) = 71.9 / 383.9$	0.225 m² (w/ margin)	✓ Done
Thermal → ρ_eff (Orbit)	$C_r = 1 + \rho_{weighted} = 1.820$ (area-weighted)	C_r = 1.820	✓ Done
Orbit → ΔV	$F_{SRP} = (S_0/c) \times A_{projected} \times C_r \rightarrow$ CR3BP integration	TBD (Member A)	⚠ Pending
Propulsion → M_prop	$m_{s/c} \times (1 - e^{(-\Delta V / Isp \times g_0)})$	TBD (Member B)	⚠ Pending
Propulsion → V_prop	m_{prop} / ρ_{iodine} (ρ = 4940 kg/m³)	TBD (Member B)	⚠ Pending

